

Pathways

Music Discovery as Exploration Rather Than Recommendation

The Problem

We live in a time of unlimited access to music.

Streaming services provide access to millions of songs and artists, yet many music enthusiasts feel that discovering new music in a meaningful way has become increasingly difficult.

Most modern recommendation systems are designed to optimize the probability of playback.

If you listen to a particular artist, you are recommended similar artists.

If you listen to a specific genre, you receive more of the same genre.

The result is often a self-reinforcing process:

- Discover Weekly gradually becomes more predictable.
- Algorithms learn the user's existing preferences but rarely challenge them.
- Music discovery becomes music optimization.

At the same time, excellent music databases and communities exist, such as Rate Your Music, Discogs, and MusicBrainz. However, these platforms primarily function as catalogues and collections of information.

What is missing is a tool where exploration itself is the goal.

Vision

Pathways is a music discovery tool that treats music as a landscape rather than a collection of categories.

Instead of asking:

"What song should I listen to next?"

Pathways asks:

"Where can I move from here?"

The goal is not to find the most likely recommendation.

The goal is to create the moments many music lovers recognize:

- The song you play five times in a row.
 - The unknown artist you cannot believe you have never discovered before.
 - The unexpected musical connection that suddenly makes sense.
 - The feeling of discovery that was often strongest in youth, when music became closely connected to identity, memories, and emotions.
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Fundamental Assumption

Traditional music systems organize music through categories.

Rock. Techno. Jazz. Post-punk.

However, people rarely experience music through categories.

When we talk about music that has mattered to us, we rarely talk about genre.

We talk about:

- Moods
- Energy
- Emotions
- Memories
- Periods of life
- Surprises
- Experiences

Two people can love the same artist for completely different reasons.

Likewise, one person can love two artists who, on paper, belong to completely different genres.

Pathways is therefore built on a different assumption:

Musical relationships are more important than musical categories.

How Does Pathways Work?

Pathways does not attempt to organize music primarily through genres.

Instead, the system starts from measurable characteristics found within the music itself.

Data from Spotify and similar sources contains a range of audio features:

- Energy
- Valence
- Danceability
- Acousticness
- Instrumentalness
- Speechiness
- Tempo

These characteristics function as coordinates in a musical space.

Each song becomes a point within this space.

Songs with similar characteristics are positioned closer together.

Songs with very different characteristics are positioned further apart.

Artists can then be represented as clusters of songs within this landscape.

The result is not a genre map.

It is a musical terrain.

A landscape of moods, energies, tensions, and possibilities.

Music as a Landscape

Imagine starting from a song that has had a significant impact on you.

Not necessarily because it belongs to a specific genre.

But because there is something about it.

A mood.

An energy.

An expression.

Pathways allows you to explore nearby areas of the musical landscape.

Not by showing "more of the same."

But by showing possible paths forward.

Some connections will feel obvious.

Others will be surprising.

The goal is not to lead the user to one correct destination.

The goal is to create interesting journeys.

Dynamic Exploration

One of the most important design principles is that exploration should be different every time.

If a user starts from the same artist on two different days, they should be able to end up in different places.

Pathways does not attempt to provide the optimal recommendation.

Pathways attempts to create new possibilities for exploration.

There is no single correct route through the musical landscape.

There are thousands.

Personal Musical Map

When users connect their Spotify account or import their playlists, the network becomes personal.

The musical landscape remains the same.

But the map looks different for every user.

For example, nodes can be visualized according to the user's relationship with them:

- Green: frequently listened to
- Yellow: familiar
- Orange: partially known
- Red: undiscovered

Pathways is therefore not only a map of music.

It becomes a map of the user's musical world.

A visual representation of the familiar, the forgotten, and the unexplored.

Interaction

Artists are visualized as nodes in a dynamic network.

Users can:

- Zoom
- Pan
- Explore connections
- Filter areas of the network
- Open detailed information about artists

When an artist is selected, an information panel can display:

- Artist information
- Top tracks
- Related areas of the network
- The user's own listening history

The graph remains simple.

The complexity lies within the relationships that can be explored.

MVP

The first version of Pathways will not attempt to directly model emotions.

Instead, measurable musical characteristics will be used as a practical proxy for musical experience.

The MVP can be built using:

- Spotify data
- Exportify data
- Audio features
- Popularity metrics
- Release years
- Artist information

The technical foundation can be built with:

- ASP.NET
- Angular
- PostgreSQL
- Sigma.js

The goal is to prove one thing:

Can we create an experience where users discover music they would not have found otherwise?

Future Possibilities

Once the foundation is working, additional layers can be added:

- Artist relationships
- Labels
- Collaborations
- User history
- Community data
- Cultural connections
- Concert data
- Other forms of art

The same exploration engine could potentially be applied to:

- Films
- TV series
- Podcasts
- Literature
- Visual art

However, music is the natural starting point because the available data is strong and the need for exploration already exists.

Core Idea

Spotify helps you listen to more music.

Pathways helps you discover more music.

Not through lists.

Not through feeds.

Not through genres.

But through exploration.